Analyzing NBA Data with GCP

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Agenda:

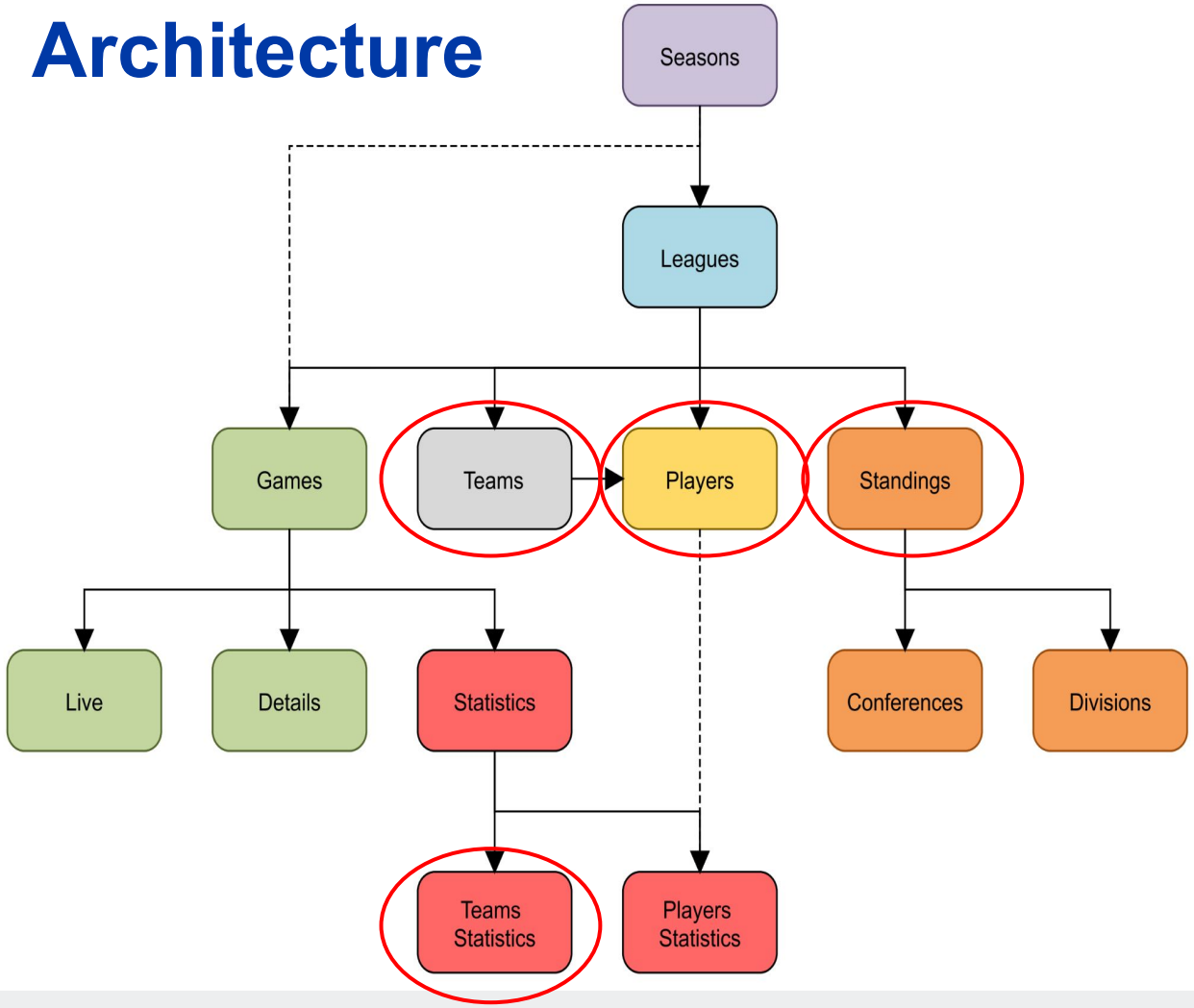
- Motivation and API Overview
- Data Extraction, Transformation, and Loading
- Data Validation
- Analysis with Queries
- Visualizations with Looker Studio
- Conclusion

Motivation & API Overview

NBA Statistics

- **Dataset**
  - **API-NBA from Rapid API**
  - **2020 season**
  - **Focusing on 4 aspects:**
    - Teams
    - Teams Statistics
    - Players
    - Standings
- **Motivation**
  - **Gain insights into relationships between different teams and their players.**
  - **Discover what makes a winning team.**

Architecture



Data Extraction, Transformation, and Loading

Data Extraction to BigQuery

- **Our Process**
  - First we defined 4 different functions to ease the process of requesting the data, creating the table, then loading that data into the created table.
- **fetch_api_data & fetch_api_data_with_param:**
  - Both functions serve the same purpose to request the data and return the response.
  - Only difference is we needed to create a “with_param” version since some of the Rapid-API calls required a specific id or team per request.
- **create_table**
  - As the name suggests, is used to create the desired BigQuery table.
- **load_data_to_bigquery**
  - This function serves as the final step to the data pipeline. Using a converted dataframe to dictionary, we pass that as “records” then load in all the data to the BigQuery table.

```python
def fetch_api_data(api_url, headers):
    response = requests.get(api_url, headers=headers)
    if response.status_code == 200:
        return response.json()
    else:
        raise Exception(f"Failed to fetch data: {response.status_code}")

def fetch_api_data_with_param(api_url, headers, param):
    response = requests.get(api_url, headers=headers, params=param)
    if response.status_code == 200:
        return response.json()
    else:
        raise Exception(f"Failed to fetch data: {response.status_code}")
```

```python
def create_table(dataset_id, table_id, credentials_path, schema):
  credentials = service_account.Credentials.from_service_account_file(credentials_path)
  client = bigquery.Client(credentials=credentials, project=[redacted])

  dataset_ref = client.dataset(dataset_id)
  table_ref = dataset_ref.table(table_id)

  try:
      client.get_table(table_ref)
      print(f"Table {dataset_id}.{table_id} already exists.")
  except NotFound:
      table = bigquery.Table(table_ref, schema=schema)
      client.create_table(table)
      print(f"Table {dataset_id}.{table_id} created successfully in BigQuery.")
```

```python
def load_data_to_bigquery(records, dataset_id, table_id, credentials_path):
    credentials = service_account.Credentials.from_service_account_file(credentials_path)
    client = bigquery.Client(credentials=credentials, project=[redacted])
    table_ref = client.dataset(dataset_id).table(table_id)
    job = client.load_table_from_json(records, table_ref)
    job.result()
    print(f"Loaded {len(records)} records into {dataset_id}.{table_id}")
```

Example - Extraction and Transformation

```python
counter = 1
skip_numbers = {3, 12, 13, 18, 32, 33, 34, 35, 36, 37, 39}
rpm = 10
delay = 60/rpm
players_list = []
api_url = "https://api-nba-v1.p.rapidapi.com/players"

while counter < 42:
  if counter in skip_numbers:
    counter += 1
    continue

  query_string = {"team":str(counter),"season":"2020"}
  player_data = fetch_api_data_with_param(api_url,headers,query_string)
  players = player_data["response"]

  for player in players:
      flattened_player = {
          'team_id': counter,
          'player_id': player['id'],
          'first_name': player['firstname'],
          'last_name': player['lastname'],
          'birthdate': player["birth"]['date'],
          'country': player["birth"]['country'],
          'rookie_year': player["nba"]['start'],
          'height_feet': player["height"]['feets'],
          'height_inches': player["height"]['inches'],
          'height_meters': player["height"]['meters'],
          'weight': player["weight"]['pounds'],
          'college': player['college'],
          'jersey_number': player["leagues"]['standard']["jersey"],
          'position': player["leagues"]['standard']["pos"],
          'is_active': player["leagues"]['standard']["active"]
      }
      players_list.append(flattened_player)

  counter += 1
  time.sleep(delay)


df_players = pd.DataFrame(players_list)
```

```python
df_players['jersey_number'] = pd.to_numeric(df_players['jersey_number'], errors='coerce')
df_players['jersey_number'] = df_players['jersey_number'].fillna(0)
df_players['jersey_number'] = df_players['jersey_number'].astype('int')

df_players['height_feet'] = pd.to_numeric(df_players['height_feet'], errors='coerce')
df_players['height_inches'] = pd.to_numeric(df_players['height_inches'], errors='coerce')
df_players['weight'] = pd.to_numeric(df_players['weight'], errors='coerce')

df_players['height_meters'] = pd.to_numeric(df_players['height_meters'], errors='coerce')
```

Example - Table Creation and Loading Data to Table

```
create_table("nba_info", "Team_Standings", [redacted], standings_schema)
```

```
Table nba_info.Team_Standings created successfully in BigQuery.
```

```
records = df_standings.to_dict(orient="records")
load_data_to_bigquery(records,"nba_info", "Team_Standings", [redacted])
```

```
Loaded 30 records into nba_info.Team_Standings
```

"Credentials" is a JSON key from a created service account that allows us to read, write, and modify tables and information in BigQuery.

Data Validation

Data Validation

- **Checking data types**
  - Defining these preset schemas, we can create tables and enforce data types. (1)

- **Null values**
  - By either setting mode to REQUIRED or NULLABLE will allow if a column can contain nulls. Throughout our tables only the id fields have been set to REQUIRED. (2)

1

```
players_schema = [
    bigquery.SchemaField("team_id", "INTEGER", mode="REQUIRED"),
    bigquery.SchemaField("player_id", "INTEGER", mode="REQUIRED"),
    bigquery.SchemaField("first_name", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("last_name", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("birthdate", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("country", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("rookie_year", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("height_feet", "FLOAT", mode="NULLABLE"),
    bigquery.SchemaField("height_inches", "FLOAT", mode="NULLABLE"),
    bigquery.SchemaField("height_meters", "FLOAT", mode="NULLABLE"),
    bigquery.SchemaField("weight", "FLOAT", mode="NULLABLE"),
    bigquery.SchemaField("college", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("jersey_number", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("position", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("is_active", "BOOLEAN", mode="NULLABLE")
]
```

```
standings_schema = [
    bigquery.SchemaField("id", "INTEGER", mode="REQUIRED"),
    bigquery.SchemaField("name", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("conference", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("conference_rank", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("division", "STRING", mode="NULLABLE"),
    bigquery.SchemaField("division_rank", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("total_wins", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("total_losses", "INTEGER", mode="NULLABLE"),
    bigquery.SchemaField("win_percentage", "FLOAT", mode="NULLABLE"),
    bigquery.SchemaField("games_behind", "FLOAT", mode="NULLABLE")
]
```

2

Analysis with Queries

Query #1 - Does Height/Weight Correlate with Wins?

- **Avg height/weight by team, sorted by win percentage**

```
SELECT
    t.name AS team_name,
    round(AVG(p.height_feet + p.height_inches / 12.0),2) AS avg_height_feet,
    round(AVG(p.weight),2) AS avg_weight_lbs,
    tstand.win_percentage
FROM nba_info.Player_Info p
JOIN nba_info.Team_Standings tstand
ON p.team_id = tstand.id
JOIN nba_info.Teams t ON p.team_id = t.id
GROUP BY t.name, tstand.win_percentage
ORDER BY tstand.win_percentage DESC;
```

Row	team_name	avg_height_feet	avg_weight_lbs	win_percentage
1	Utah Jazz	6.51	225.07	0.722
2	Phoenix Suns	6.49	211.43	0.708
3	Philadelphia 76ers	6.51	216.81	0.681
4	Brooklyn Nets	6.53	219.43	0.667
5	Denver Nuggets	6.53	220.0	0.653
6	LA Clippers	6.45	213.2	0.653
7	Milwaukee Bucks	6.49	220.19	0.639
8	Dallas Mavericks	6.56	218.0	0.583
9	Los Angeles Lak...	6.58	229.75	0.583
10	Portland Trail Bla...	6.51	215.87	0.583

Query #2 - Do Second Chance Points / Points From Turnovers Correlate with Wins?

- **Second chance points & points off turnovers of teams sorted by win percentage**
  - SELECT
  - t.name AS team_name,
  - tstats.second_chance_points,
  - tstats.points_off_turnovers,
  - tstand.win_percentage
  - FROM nba_info.Team_Statistics tstats
  - JOIN nba_info.Team_Standings tstand ON tstats.id = tstand.id
  - JOIN nba_info.Teams t ON tstats.id = t.id
  - ORDER BY tstand.win_percentage desc;

Row	team_name	second_chance_poin	points_off_turnovers	win_percentage
1	Utah Jazz	1173	1219	0.722
2	Phoenix Suns	1188	1569	0.708
3	Philadelphia 76ers	1172	1511	0.681
4	Brooklyn Nets	1031	1375	0.667
5	LA Clippers	1156	1462	0.653

Row	team_name	second_chance_poin	points_off_turnovers	win_percentage
1	Houston Rockets	928	1388	0.236
2	Detroit Pistons	943	1262	0.278
3	Orlando Magic	1007	1056	0.292
4	Oklahoma City Thunder	866	1075	0.306
5	Cleveland Cavaliers	973	1251	0.306

Query #3 - Do turnovers have any relation with total losses?

- **Turnovers of teams sorted by total losses**
  - SELECT
  - t.name AS team_name,
  - tstats.turnovers,
  - tstand.total_losses
  - FROM nba_info.Team_Statistics tstats
  - JOIN nba_info.Team_Standings tstand ON tstats.id = tstand.id
  - JOIN nba_info.Teams t ON tstats.id = t.id
  - ORDER BY tstand.total_losses desc;

Row	team_name	turnovers	total_losses
1	Utah Jazz	1154	20
2	Phoenix Suns	1182	21
3	Philadelphia 76ers	1177	23
4	Brooklyn Nets	1109	24
5	LA Clippers	1126	25

Row	team_name	turnovers	total_losses
1	Houston Rockets	1075	55
2	Detroit Pistons	1103	52
3	Orlando Magic	941	51
4	Oklahoma City Thunder	1161	50
5	Cleveland Cavaliers	1131	50

Query #4 - Identifying New Talent

- **List players who debuted after 2019 and belong to a team with win percentage > 0.6**

```
SELECT
    p.first_name,
    p.last_name,
    p.rookie_year,
    t.name AS team_name,
    tstand.win_percentage
FROM nba_info.Player_Info p
JOIN nba_info.Team_Standings tstand ON p.team_id = tstand.id
JOIN nba_info.Teams t ON p.team_id = t.id
WHERE p.rookie_year > 2019
    AND tstand.win_percentage > 0.6
ORDER BY tstand.win_percentage DESC, p.rookie_year ASC;
```

Row	first_name	last_name	rookie_year	team_name	win_percentage
1	Trent	Forrest	2020	Utah Jazz	0.722
2	Elijah	Hughes	2020	Utah Jazz	0.722
3	Udoka	Azubuike	2020	Utah Jazz	0.722
4	Trevon	Scott	2021	Utah Jazz	0.722
5	Ty-Shon	Alexander	2020	Phoenix Suns	0.708

Query #5 - Summary Statistics

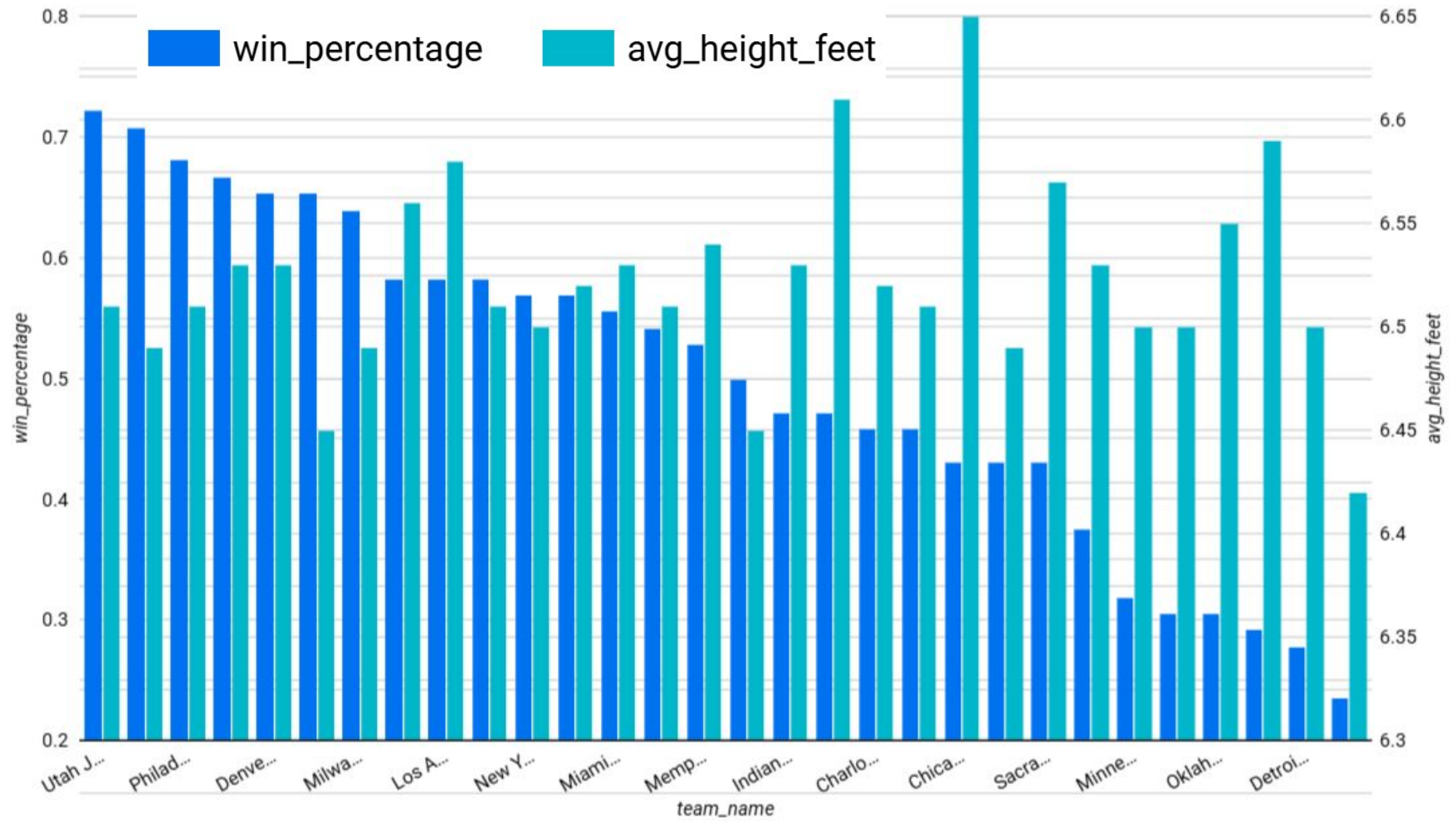
- **Average team statistics grouped by division**
  - SELECT
  - tstand.division,
  - AVG(tstat.tot_reb) AS avg_rebound,
  - AVG(tstat.assists) AS avg_assists,
  - AVG(tstat.blocks) AS avg_blocks,
  - AVG(tstat.steals) AS avg_steals
  - FROM nba_info.Team_Statistics tstat
  - JOIN nba_info.Teams t ON tstat.id = t.id
  - JOIN nba_info.Team_Standings tstand on t.id = tstand.id
  - GROUP BY tstand.division;

Row	division	avg_rebound	avg_assists	avg_blocks	avg_steals
1	southeast	3611.6	2007.6	359.0	599.6
2	northwest	3644.8	1915.0	394.0	594.6
3	southwest	3503.4	1923.4	372.8	589.0
4	atlantic	3596.8	1938.8	444.6	641.4
5	central	3598.0	2038.4	395.4	622.6
6	pacific	3693.6	2161.4	399.4	642.8

Visualizations with Looker Studio

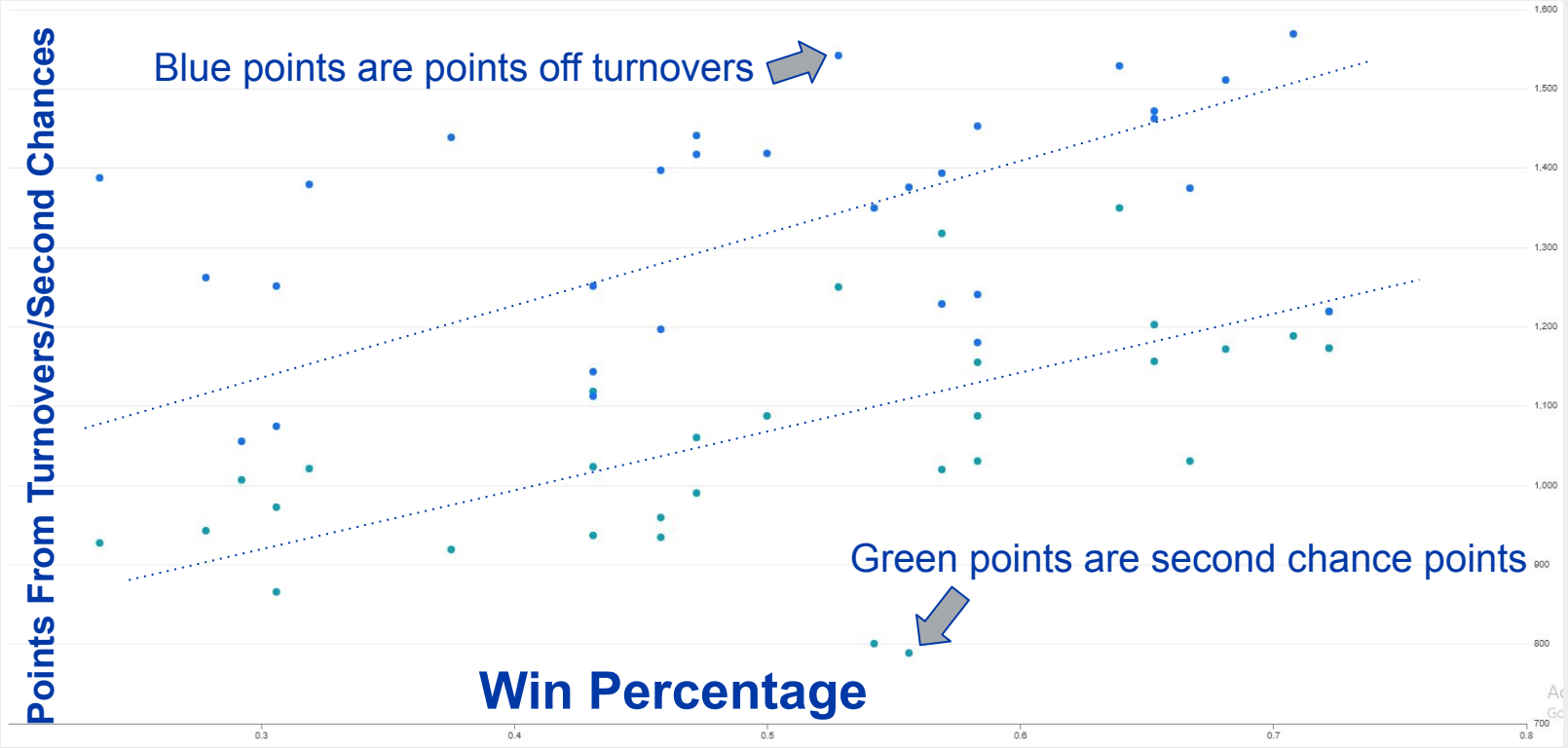
Query #1 - Does Height Correlate with Wins?

- **Insights**
  - Average height beyond 6.5 ft. does not seem to indicate a winning team



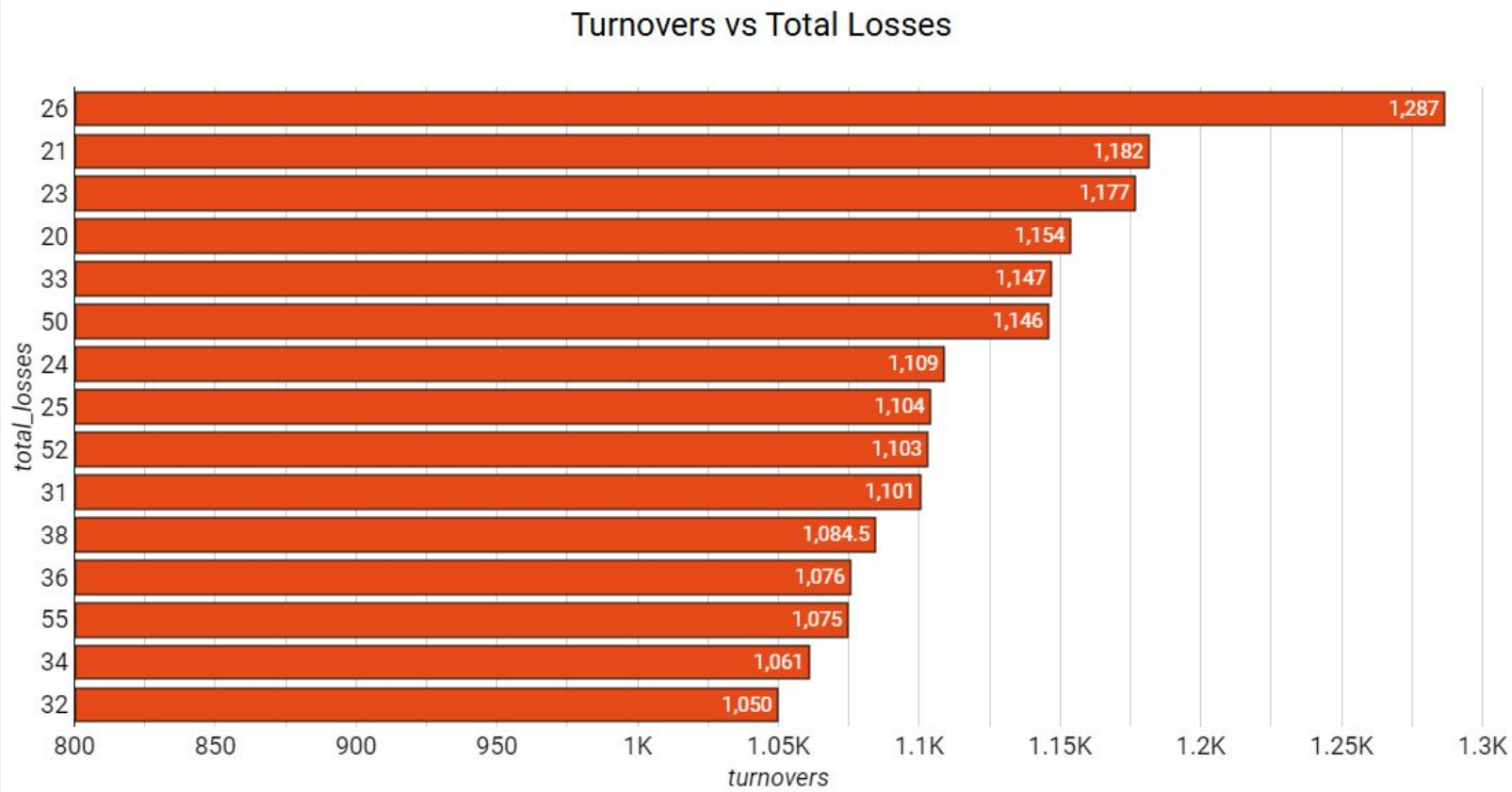
Query #2 - Do Second Chance Points / Points From Turnovers Correlate with Wins?

- **Insights**
  - Winning teams tend to have more points off turnovers and second points, as expected
  - This plot shows how important offensive rebounds can be and capitalizing on turnovers to generate points



Query #3 - How turnovers effect losses

- **Insights**
  - Surprisingly, even though points off turnovers seems to have a positive relationship with win percentage, we observe that turnovers alone don't have any relationship with losses.
  - This shows how important it is to get back on defense after a turnover.



Conclusion

Key Findings

- **Height is not a clear indicator of team success**
- **Points from turnovers & second chances correlate positively with win percentage**
- **The Pacific division had the strongest performance on key metrics during the 2020 season**

Row	division	avg_rebound	avg_assists	avg_blocks	avg_steals
1	southeast	3611.6	2007.6	359.0	599.6
2	northwest	3644.8	1915.0	394.0	594.6
3	southwest	3503.4	1923.4	372.8	589.0
4	atlantic	3596.8	1938.8	444.6	641.4
5	central	3598.0	2038.4	395.4	622.6
6	pacific	3693.6	2161.4	399.4	642.8

2020 NBA Champions

Thank you